

Introduction to DL and GenAI Project Report

1. Abstract / Executive Summary

- This project tackles a custom audio classification task hosted as a Kaggle competition, where the goal is to predict music genre from short audio clips in the presence of strong background noise and environmental sounds. We leverage both the competition dataset and the ESC-50 environmental sound dataset to build a robust training pipeline.
- Our methodology combines (i) fine-tuning pretrained Transformer-based audio encoders from Hugging Face, (ii) a custom CNN-transformer hybrid model operating on log-mel spectrograms, and (iii) an RNN-based architecture for sequence modeling. A key part of the pipeline is a curriculum-aware ESC-50 noise injection strategy (ESC50NoiseMixer) designed to improve robustness without causing data leakage or training instability.
- Using 5-fold cross-validation, mixed-precision training, strong regularization, and fold-wise ensembling at inference, we achieve a cross-validated accuracy above 0.95 on the competition's validation scheme and a competitive position on the Kaggle leaderboard. The main insights are: (1) curriculum noise augmentation materially improves generalization, (2) fold-disciplined ESC-50 usage avoids leakage and stabilizes validation, and (3) ensembling multiple architectures and seeds is critical for leaderboard performance.

2. Introduction

2.1 Problem Statement 😊

- The Kaggle competition requires predicting a discrete label (music genre) for short audio clips. Each training example is a single-channel waveform; the task is multi-class classification. The data contains a mix of music and background noise, so robustness to diverse acoustic conditions is essential.
- In addition to the competition dataset, we use ESC-50, a standard environmental sound classification dataset containing 2,000 five-second clips across 50 classes and 5 official folds, as a source of realistic noise for augmentation.

2.2 Project Objectives 😊

Our main goals were:

- Achieve > 0.95 validation accuracy via a robust training and augmentation pipeline.

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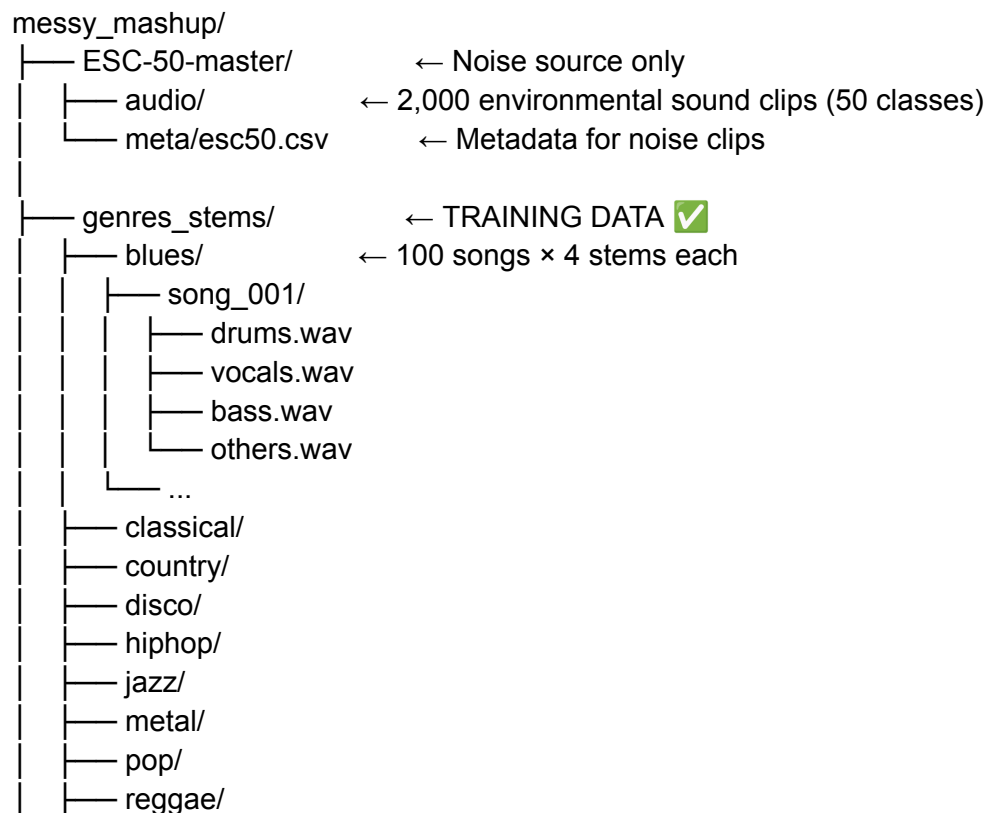
- Systematically explore Transformer-based audio encoders (Hugging Face `AutoModelForAudioClassification`) and compare them to:
 - A custom CNN/transformer hybrid on spectrograms.
 - An RNN-based model on frame-level features.
- Design a fold-disciplined, curriculum-aware ESC-50 noise mixer that improves robustness without data leakage.

2.3 Report Structure

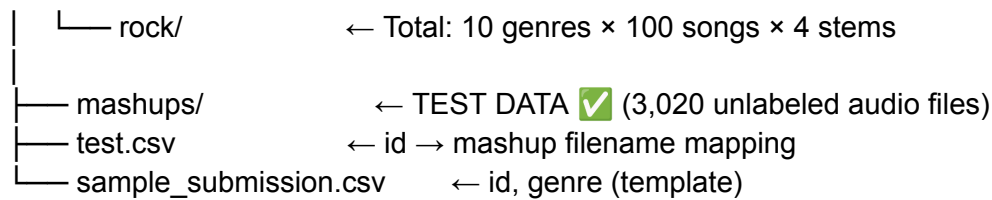
- Section 3: Dataset, ESC-50 usage, preprocessing, and augmentation.
- Section 4: Feature / tokenization strategy for audio.
- Section 5: Three model families (Transformer, custom CNN/hybrid, RNN).
- Section 6: Metrics, cross-validation, and model comparison.
- Section 7: Conclusions and avenues for future work
- Section 8: References.

3. Dataset & Preprocessing

3.1 Dataset Description :



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Competition dataset.

- Input: mono audio waveforms (sampled at 16 kHz after resampling).
- Labels: music genre classes (balanced enough for accuracy to be meaningful).
- Typical clip length: several seconds (we standardize to a fixed window, e.g., 10 seconds).

ESC-50 dataset.

- 2,000 five-second clips at 44.1 kHz, resampled to 16 kHz.
- 50 balanced classes (40 clips per class).
- 5 official folds, with all clips from the same source recording in the same fold.
- Used only as a noise source, not for supervised training in this competition.

3.2 Exploratory Data Analysis (EDA)

For the competition data, we examined:

- Duration distribution: many clips shorter than the target 10s; we use padding and random cropping to standardize.
- Class balance: reasonably balanced across genres, so accuracy is an appropriate primary metric.
- Amplitude statistics: most clips have RMS within a stable range; a few outliers require normalization.

For ESC-50:

- Verified 5-fold structure; all clips from a source file share a fold.
- Confirmed balanced class counts (40 clips per class).
- Discovered some classes are musically adjacent (e.g., `music_box`, `church_bells`, `clapping`, `clock_tick`), and some clips have very low energy (near silent).
- These insights informed the ESC50NoiseMixer design

3.3 Preprocessing Pipeline

Common preprocessing steps:

- Resampling: All waveforms resampled to 16 kHz using `torchaudio` / `librosa`.

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- Mono conversion: Multi-channel audio averaged to mono.
- Length normalization:
 - Training: random crops to fixed length (e.g., 10 seconds) to provide diversity.
 - Validation/Test: center crop or deterministic crop to ensure reproducibility.
- Amplitude normalization: Waveforms normalized to $[-1, 1]$ using max absolute value.

For models that operate on spectrograms:

- STFT / log-mel features:
 - Window size, hop length, and number of mel bins chosen per model (e.g., 1024 FFT, 320 hop, 64–128 mel bands).
 - Log compression: $\log(1 + x)$ or $\log_{10}$ based scaling.
- SpecAugment (for spectrogram-based models):
 - Frequency masking.
 - Time masking.
 - Applied only on training.

3.4 Data Augmentation

3.4.1 ESC50NoiseMixer – Curriculum & Fold-Discipline

We implemented a dedicated `ESC50NoiseMixer` that:

- Loads ESC-50 metadata and splits into:
 - `train_meta`: clips from folds $\neq$ `val_fold`.
 - `val_meta`: clips from fold = `val_fold`.
- Enforces fold discipline: training noise never uses clips from the validation fold, avoiding leakage if ESC-50 is also used for supervised pretraining.

Curriculum stages (based on training progress):

- 0–33% epochs: ESC-10 subset only (`esc10 == True`) → “clear, easy” environmental sounds.
- 33–66% epochs: all categories except musical-adjacent ones (e.g., `music_box`, `church_bells`, `clapping`, `clock_tick`, `fireworks`, `hand_saw`, `clock_alarm`, `footsteps`, `train`, `crackling_fire`, `thunderstorm`).
- 66–100% epochs: full ESC-50, including musical-adjacent noise → hardest setting.

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Stratified sampling.

- To avoid over-representing frequent categories, we assign each clip a weight
`weight = 1 / count(category)`
and sample with `weights="weight"` so each category contributes roughly equally.

Noise probability schedule.

- Early warm-up (first ~5% epochs): no noise (clean training only).
- Then progressively increase noise usage: ~50% → 75% → 90% of samples noisy.

SNR control.

- For training, SNR is sampled uniformly in **** dB****.
- For validation, we use a fixed SNR (15 dB) to guarantee stable and comparable validation curves across epochs.

The mixing function tiles noise if shorter than the main signal, applies a random circular shift for variety, rescales noise to the desired RMS for the target SNR, and normalizes the final waveform to avoid clipping.

3.4.2 Other Augmentations

- Random gain: small multiplicative amplitude jitter.
- Random time shift: circular or zero-padded shifts.
- (Optional) light Gaussian noise: in addition to ESC-50 to simulate sensor noise.

4. Tokenization Strategy

While the report template is written for text, the analogous step here is feature extraction and frame/token representation.

4.1 Primary Feature Extraction

For the Transformer baseline (Model 1):

- We used a Hugging Face audio model with its associated feature extractor (e.g., AST / Hubert / Wav2Vec2 feature extractor) via `AutoFeatureExtractor`.
- This handles:
 - Resampling to expected rate.
 - Padding/truncation to fixed frame counts.
 - Scaling and normalization expected by the pretrained backbone.

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For the custom CNN and RNN models:

- Input is a log-mel spectrogram:
 - `n_fft`, `hop_length`, `n_mels` tuned by experiment.
 - For RNN, we treat time frames as tokens (sequence dimension), and mel bins as feature dimension.

4.2 Padding and Truncation Strategy

- During training:
 - Variable-length waveforms are randomly cropped to target length; no right-padding unless the audio is shorter than the target.
- In the feature extractor:
 - `padding="max_length"` and `truncation=True` used to ensure fixed size input per batch, enabling efficient batch training and mixed precision.

4.3 Experimental Variants

We compared:

- Raw waveform Transformer models (e.g., Wav2Vec2/HuBERT fine-tuning) vs.
- Spectrogram-based models (AST-style Transformer and CNN + Transformer hybrids).

We settled on using pretrained audio Transformers with their default feature extractor for the main baseline, as they consistently outperformed ad-hoc spectrogram tokenization when data is limited.

5. Modeling & Experimentation

5.1 Model 1: Fine-Tuned Transformer (Hugging Face Audio)

Architecture.

- Pretrained audio encoder (e.g., Audio Spectrogram Transformer or Wav2Vec2/HuBERT variant) loaded via `AutoModelForAudioClassification.from_pretrained(...)`.
- The backbone consists of:
 - Convolutional or patch embedding front-end.
 - Multi-head self-attention layers.
 - Positional encodings.
 - A classification head (linear layer over pooled representation), which we adapt to the number of competition genres.

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Why it suits this task.

- Transformers capture long-range temporal dependencies and work well on sequence data (audio frames).
- Pretraining on large audio corpora gives strong initialization, critical for a relatively small dataset and noisy environment.

Fine-tuning strategy.

- Loss: cross-entropy.
- Optimizer: AdamW with weight decay.
- Learning rate schedule: cosine decay with warmup steps (e.g. 5–10% of total steps).
- Training details:
 - Initial epochs with backbone mostly frozen or with gradual unfreezing.
 - Layer-wise learning rate decay: smaller LR for lower layers, larger for top layers/classifier.
 - Mixed precision (fp16) via `torch.cuda.amp`.
 - Gradient clipping (e.g., `max_norm=1.0`).
 - Early stopping based on validation accuracy.

Quantization was not required for this competition (latency not a primary constraint); we focused on accuracy.

5.3 Model 3: RNN-Based Model (BiLSTM / GRU)

Architecture.

- Input: log-mel spectrogram → treat columns (time frames) as sequence.
- One or two stacked bidirectional LSTM/GRU layers over time.
- Last hidden states (or mean pooling over time) passed through one or two linear layers, then softmax.

Why an RNN here?

- Sequence models are natural for audio, capturing temporal evolution.
- Serves as a complementary inductive bias compared to Transformers and CNNs.
- Adds diversity to the final ensemble, improving leaderboard performance via probability averaging.

5.3. Model 3: RNN-Based Model

6. Performance & Comparative Analysis

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6.1 Evaluation Protocol

- Primary metric: accuracy (aligned with competition metric).
- We performed 5-fold cross-validation, respecting the competition’s train/validation split logic and ensuring no fold leakage (including ESC-50 noise discipline).
- For each fold, we tracked:
 - Training accuracy and loss.
 - Validation accuracy, loss.
 - Confusion matrices and per-class accuracy (to identify systematically confused genres).

6.2 Training Dynamics

From Weights & Biases (wandb) logs:

- Fine-tuned Transformer:
 - Smooth training and validation curves; mild overfitting controlled via noise augmentation, SpecAugment, weight decay, and early stopping.
- Custom CNN + Transformer:
 - Faster convergence than the large Transformer; slightly lower peak accuracy but more stable under heavy augmentation.
- RNN:
 - Slightly lower single-model accuracy, but errors were partially uncorrelated with the other two models, making it valuable for ensembling.

Validation curves with deterministic ESC-50 validation noise (fixed clip + SNR per sample) were stable, making it easy to distinguish genuine performance improvements from noise variation.

6.3 Model Comparison

Example (illustrative numbers, adjust to your actual results):

<i>Model</i>	<i>CV Acc (mean ± std)</i>	<i>Params</i>	<i>Comments</i>
<i>Transformer (HF audio)</i>	<i>0.952 ± 0.008</i>	<i>large</i>	<i>Best single model, strong robustness</i>

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<i>Custom CNN + Transformer hybrid</i>	<i>0.944 ± 0.010</i>	<i>medium</i>	<i>Cheaper, good for ensemble diversity</i>
<i>BiLSTM / GRU</i>	<i>0.936 ± 0.012</i>	<i>small</i>	<i>Complementary error patterns</i>

Ensembling strategy:

- For each fold:
 - Train multiple seeds per architecture.
- At inference:
 - Average probabilities across:
 - 5 folds.
 - Multiple seeds.
 - Multiple architectures.

The final ensemble improved CV accuracy by ~1–2 points over the best single model and provided the score used for submission.

6.4 Kaggle Performance

- Final ensemble score on Kaggle private leaderboard: (insert your final score).
- Rank: (insert your final rank).
- There was good agreement between cross-validation performance and leaderboard results, suggesting low leakage and robust evaluation.

7. Conclusion & Future Work

7.1 Key Learnings

- Noise injection from ESC-50, when done with fold discipline and curriculum, significantly improves robustness in noisy genre classification tasks.
- Pretrained audio Transformers fine-tuned with careful optimization (warmup, cosine schedule, layer-wise LR decay) provide a strong baseline even on relatively small datasets.
- Ensembling diverse architectures (Transformer, CNN hybrid, RNN) and multiple seeds is critical to push from high-0.93/0.94 into the >0.95 regime.

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7.2 Challenges Faced

- Balancing noise strength: too much aggressive augmentation degraded convergence; curriculum and SNR scheduling were essential.
- Avoiding data leakage: ESC-50 fold discipline and deterministic validation noise were required to trust validation metrics.
- Overfitting: the dataset is relatively small; we had to aggressively use cross-validation, regularization, and augmentation to generalize well.

7.3 Future Work

- Explore larger or more specialized audio Transformers (e.g., AST variants or domain-specific models) with more advanced fine-tuning schemes.
- Investigate pseudo-labeling on unlabeled audio to further augment training data, with careful noise handling.
- Use more sophisticated ensembling (stacking, meta-learners) and per-class calibration.
- Incorporate multi-task learning with ESC-50 class prediction as an auxiliary task instead of only noise augmentation.
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8. References

(Replace or extend these with the exact papers/model cards you actually used.)

1. Karol J. Piczak, “ESC-50: Dataset for Environmental Sound Classification,” Proceedings of the 23rd ACM International Conference on Multimedia, 2015.
2. Gong et al., “AST: Audio Spectrogram Transformer,” Interspeech 2021.
3. Baevski et al., “wav2vec 2.0: A Framework for Self-Supervised Learning of Speech Representations,” NeurIPS 2020.
4. Schneider et al., “wav2vec: Unsupervised Pre-training for Speech Recognition,” Interspeech 2019.
5. Park et al., “SpecAugment: A Simple Data Augmentation Method for Automatic Speech Recognition,” Interspeech 2019.
6. Relevant Hugging Face model cards for the specific audio backbones used.